B. Tech Degree VI Semester (Supplementary) Examination September 2008

ME 603 INSTRUMENTATION THEORY AND CONTROL ENGINEERING

(2002 Scheme)

Time : 3 Hours | Maximum Marks : 100

I. (a) Explain the method of High gain feed back for correcting spurious inputs. (10)
   (b) What are Back lash, Dead space and Threshold? (3)
   (c) A wheatstones' bridge requires a change of 7Ω in the unknown arm of the bridge to produce a change of 3 mm in the deflection of the galvanometer. Determine sensitivity of bridge. Also determine the deflection factor. (7)

OR

II. (a) Show that 'Mercury in a glass thermometer' can be used as a first order instrument. (12)
   (b) Obtain static sensitivity and time constant of this instrument. (5)
   (c) What are the associated assumptions? (3)

III. (a) For a strain gauge show that
   Gauge factor = 1 + 2 x poisson ratio. (12)
   (b) Find the change in nominal wire resistance of 120Ω which results from a strain of 1000 micrometre/metre. The gauge factor for the resistance wire is 2. (8)

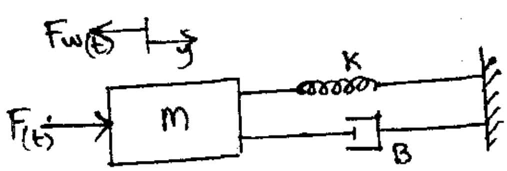
OR

IV. (a) Explain 'law of intermediate metals' and 'law of intermediate temperatures' concerning Thermo couples. (6)
   (b) With a schematic diagram explain working of Geiger Muller tube. (9)
   (c) What are different types of microphones? (5)

V. (a) Derive the differential equation describing the dynamics of liquid level system when liquid inflow rate is suddenly increased from its steady state conditions. (10)
   (b) A unit step input is applied to unity feed back system whose open loop transfer function is given by $G(s) = \dfrac{10}{s(s-11)}$. Find the natural and damped frequencies of operation. (10)

OR

VI. (a) Explain robot arm system with a suitable diagram. (10)
   (b) Obtain the transfer function of mechanical system shown below. (10)



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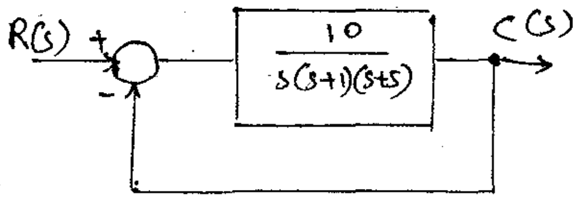
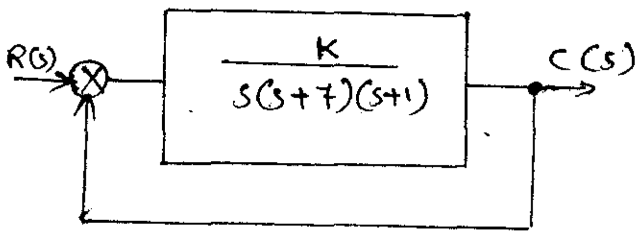
VII. (a) Use Bode plot to determine the stability of unity feed back system shown below. (10)



(b) Determine the value of K for which the system will be stable. (10)



OR

VIII. (a) Using Hurwitz criterion verify the stability of $s^4 + 8 s^3 + 18 s^2 + 16 s + 5 = 0$ (10)

(b) What is Liapanov function? How stability is analysed using Liaponov approach? (10)

IX. Write short notes on :

(i) Pneumatic controllers (ii) Servomotors

(iii) Stepper motors (iv) Integral control action. (4 x 5 = 20)

OR

X. (a) Derive the transfer function of an armature controlled DC servo motor and explain. (10)

(b) Write short notes on :

(i) Tacho generators (ii) Synchros. (2 x 5 = 10)

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